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Bisection Method
The bisection method is a simple and robust numerical technique used to find the roots of a continuous function. It works by iteratively narrowing down an interval where a root is known to exist, based on the Intermediate Value Theorem. This theorem states that if a continuous function changes sign over an interval, then there is at least one root within that interval. The method is particularly useful when an analytical solution is difficult or impossible to obtain.

Steps of the Bisection Method

1. **Initial Interval Selection**: Choose initial points a and b such that $f(a) \cdot f(b) < 0$.
2. **Midpoint Calculation**: Compute the midpoint $c = \frac{a+b}{2}$.
3. **Interval Reduction**: Determine whether the root lies in $[a, c]$ or $[c, b]$ by checking the sign of $f(c)$:
   - If $f(a) \cdot f(c) < 0$, then the root is in $[a, c]$.
   - If $f(b) \cdot f(c) < 0$, then the root is in $[c, b]$.
4. **Iteration**: Repeat the process until the interval is sufficiently small.

Problem: Solving $x^3 + 4x^2 = 10$ in the Interval $[1, 2]$
a) What is the approximation after 5 iterations?

1. **Initial interval**: $[1, 2]$
   - Compute $f(1) = 1^3 + 4(1)^2 - 10 = -5$
   - Compute $f(2) = 2^3 + 4(2)^2 - 10 = 14$
   - Since $f(1) \cdot f(2) < 0$, a root exists in $[1, 2]$.
2. **Iteration 1**:
   - $c = \frac{1+2}{2} = 1.5$
   - Compute $f(1.5) = 1.5^3 + 4(1.5)^2 - 10 = 2.375$
   - Since $f(1) \cdot f(1.5) < 0$, new interval is $[1, 1.5]$.
3. **Iteration 2**:
   - $c = \frac{1+1.5}{2} = 1.25$
   - Compute $f(1.25) = 1.25^3 + 4(1.25)^2 - 10 = -1.796875$
   - Since $f(1.25) \cdot f(1.5) < 0$, new interval is $[1.25, 1.5]$.
4. **Iteration 3**:
   - $c = \frac{1.25+1.5}{2} = 1.375$
   - Compute $f(1.375) = 1.375^3 + 4(1.375)^2 - 10 = 0.162109$
   - Since $f(1.25) \cdot f(1.375) < 0$, new interval is $[1.25, 1.375]$.
5. **Iteration 4**:
   - $c = \frac{1.25+1.375}{2} = 1.3125$
   - Compute $f(1.3125) = 1.3125^3 + 4(1.3125)^2 - 10 = -0.848388$
   - Since $f(1.3125) \cdot f(1.375) < 0$, new interval is $[1.3125, 1.375]$.
6. **Iteration 5**:
   - $c = \frac{1.3125+1.375}{2} = 1.34375$
   - Compute $f(1.34375) = 1.34375^3 + 4(1.34375)^2 - 10 = -0.350982$
   - Since $f(1.34375) \cdot f(1.375) < 0$, new interval is $[1.34375, 1.375]$.

Approximation after 5 iterations: 1.34375

b) What are the conditions for the bisection method to converge?
Conditions for Convergence:

- The function $f(x)$ must be continuous on the interval $[a, b]$.
- The initial interval $[a, b]$ must be chosen such that $f(a) \cdot f(b) < 0$, indicating that the function changes sign over the interval. This ensures there is at least one root within the interval.

c) What is the minimum number of iterations to find an approximation to the solution, using the bisection method, with an absolute error smaller than 10^{-4}?
To determine the minimum number of iterations required to achieve an absolute error smaller than 10^{-4}, we use the formula for the interval width after n iterations:

$$\text{Width after } n \text{ iterations} = \frac{b-a}{2^n}$$

Given that $b - a = 1$ (since the interval is $[1, 2]$), we want the interval width to be less than 10^{-4}:

$$\frac{1}{2^n} < 10^{-4}$$

Taking the logarithm (base 10) of both sides:

$$\log_{10}\left(\frac{1}{2^n}\right) < \log_{10}(10^{-4})$$

$$-n \log_{10}(2) < -4$$

Solving for n:

$$n > \frac{4}{\log_{10}(2)}$$

Since $\log_{10}(2) \approx 0.3010$:

$$n > \frac{4}{0.3010} \approx 13.2877$$

Therefore, the minimum number of iterations n must be: $n \geq 14$
Minimum number of iterations: 14